Strategy comparison

Strategy comparison overview Scope - This document compares all strategy implementations cur
src/strategies. - It summarizes signal style, complexity, risk model, and notable implementat
- Source details come from the per-strategy walkthrough files in docs/strategy_walkthroughs.
included 1) ema_cross 2) rsi_reversion 3) donchian_breakout 4) ema_trend_hold 5) bmsb 6)
emalyarovich_smas 7) k_davey_mom_keltner 8) basic_keltner_reversion Comparison matrix | Str
Signal style | Market bias | Shorts | Risk model | Position sizing | Complexity |
|---|---|---|---|---|---|---| | ema_cross | Trend-following EMA crossover with trend filter |
| Percent SL/TP, optional ATR SL/TP, optional ADX gate | Backtester modes + optional position
Medium | | rsi_reversion | Mean reversion from RSI oversold to RSI recovery | Mean reversion
(default) | Percent SL/TP only | Backtester modes + optional position_pct | Low | | donchian_
Breakout above prior high, exit below prior low | Trend breakout | No (default) | Percent SL/
optional ATR SL/TP | Backtester modes + optional position_pct | Low-Medium | | ema_trend_hold
above trend EMA, hold until close below trend EMA | Trend hold | No (default) | Percent SL/TP
Backtester modes + optional position_pct | Low | | bmsb | Price over blended baseline plus ro
signal confirmation | Trend with confirmation | No (default) | Optional custom trailing stop
Backtester SL/TP inputs | Backtester modes + optional position_pct | Medium | | emalyarovich_
Pullback to fast SMA inside higher-timeframe trend | Trend pullback | No (default) | Percent
Backtester modes + optional position_pct | Medium | | k_davey_mom_keltner | Momentum + trend
+ Keltner thresholds, next-bar execution | Multi-filter trend/momentum | Yes | ATR-based stop
futures mode, no fixed percent SL/TP | Dynamic contracts or optional position_pct | High | |
basic_keltner_reversion | Revert from KC extremes back toward EMA centerline | Mean reversion
Percent SL/TP only | Backtester modes + optional position_pct | Low-Medium | Step-by-step me
for reading any strategy in this repo 1) Data load and warmup - The runner pulls OHLCV with f
and computes indicators. - Warmup length is strategy-specific (EMA/RSI/Donchian/SMA/ATR lookb
Bar loop order - Most runners call bt.on_bar first each bar. - This means stop/take-profit ch
fire before new signal logic. 3) Signal generation - Strategy function returns a tuple: (sig
trigger). - Common signals are LONG, SHORT, EXIT, or None. 4) Signal execution - bt.on_signa
position transitions. - Pyramiding behavior and trigger fallback labels are handled in Backte
Outputs - Every runner computes stats, equity curve, CSV, and optional QuantStats report. - S
extra plot generation flags; behavior is not always identical across runners. Complexity ran
(highest to lowest) 1) k_davey_mom_keltner 2) ema_cross 3) bmsb 4) emalyarovich_smas 5) donch
6) basic_keltner_reversion 7) rsi_reversion 8) ema_trend_hold Why this ranking - k_davey_mom
combines multi-filter entry logic, next-bar pending actions, ATR stop updates, futures PnL, a
contract sizing. - ema_cross and bmsb add meaningful filter/trailing behavior beyond simple o
triggers. - rsi_reversion and ema_trend_hold are straightforward single-indicator systems wit
entry/exit gates. Key cross-strategy caveats you should know 1) Timing semantics are not ful
- Some runners compute signals on prior-bar slices and execute on current close; K. Davey que
for next-bar open. 2) CSV schema is reused across strategies. - Fields like ema_fast/ema_slo
even for non-EMA-crossover systems (often set to None or repurposed). 3) Repeated entry sign
happen by design. - Several strategies emit LONG repeatedly while condition stays true; actua
depend on BacktesterV2 position/pyramiding checks. 4) Exit generation differs by strategy st
exits are explicit indicator exits. - Others rely heavily on BacktesterV2 stop logic once in
Recommended usage by archetype - If you want the simplest baseline: ema_trend_hold or rsi_rev
you want breakout logic: donchian_breakout. - If you want pullback trend entries: emalyarovic
you want advanced futures-style dynamic sizing: k_davey_mom_keltner. - If you want mean-rever
channel extremes and both sides: basic_keltner_reversion.

Synthetic baseline comparison